

Problem

Euclid's Identity states that for any two positive integers A and B, there exist integers

x and y such that:  $Ax + By = D$

where D is the Greatest Common Divisor

(GCD) of A and B. Given two positive

integers A and B, find:  $x \ y \ D = \gcd(A, B)$

such that:

$Ax + By = D$

If multiple pairs (x, y) satisfy the equation,

choose the pair with the minimum value of:

$|x| + |y|$

If there is still more than one pair, choose the

pair for which:

$x \leq y$

Submissions

Leaderboard

Input Format

A B

Constraints

$1 \leq A, B \leq 10^9$

Discussions

Output Format

$x \ y \ D$  D is  $\gcd(A, B)$   $Ax + By = D$  (x, y)

satisfies the minimum-selection rules.

Sample Input 0

4 6

Sample Output 0

-1 1 2

Change Theme

Language

Python 3



```
1 A, B = map(int, input().split())
2 def egcd(a, b):
3     if b == 0:
4         return a, 1, 0
5     d, x1, y1 = egcd(b, a % b)
6     x = y1
7     y = x1 - (a // b) * y1
8     return d, x, y
9 D, x, y = egcd(A, B)
10 bx = x
11 by = y
12 k = round(-x * D / B)
13 for i in range(k - 2, k + 3):
14     nx = x + i * (B // D)
15     ny = y - i * (A // D)
16     if abs(nx) + abs(ny) < abs(bx) + abs(by):
17         bx = nx
18         by = ny
19     elif abs(nx) + abs(ny) == abs(bx) + abs(by):
20         if nx <= ny:
21             bx = nx
22             by = ny
23 print(bx, by, D)
```

Line: 22 Col: 20

Upload Code as File

☐ Test against custom input

Run Code

Submit Code

# Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Compiler Message

Success